example, the antimaterialist culture of interwar Weimar Germany likely played a role. And Heisenberg and others were definitely influenced by Ernst Mach and his successors, the “Vienna Circle” of philosophers, who developed a school of thought called “logical positivism.” Logical positivism picked up where Mach left off—according to them, any statement that made reference to something unobservable was not only bad science, it was literally meaningless. Thus, talking about what happens in quantum systems when nobody’s looking is nonsensical.

The influence of the logical positivists on the founders of quantum physics was particularly personal in the case of Wolfgang Pauli. Pauli was born and raised in Vienna, and his godfather was Ernst Mach himself. Outspoken, quick-witted, and deeply talented, Pauli had enormous influence among the physicists of his day. Heisenberg and Bohr both craved his good opinion. But it was hard to come by—Pauli’s scathing put-downs were legendary, earning him the nickname “the Wrath of God.” “I do not mind if you think slowly, but I do object when you publish more quickly than you can think,” he once told a colleague. He dismissively said of another physicist’s paper that “it is not even wrong.” Even his compliments were backhanded: after hearing Einstein lecture to a full house at the University of Munich, Pauli exclaimed, “You know, what Mr. Einstein said is not so stupid.” And, when discussing matters of quantum interpretation, Pauli often sounded a positivist note. According to him, worrying about the position of an object before it’s been measured was pointless. “One should no more rack one’s brain about the problem of whether something one cannot know anything about exists all the same,” he said, “than about the ancient question of how many angels are able to sit on the point of a needle.”

Positivism influenced the rest of the Copenhagen camp, but to varying degrees. And they applied it in different ways, leading to inconsistent views among them. Bohr simply dismissed the idea of a quantum world altogether. “There is no quantum world,” he said. “Isolated material particles are abstractions, their properties on the quantum theory being definable and observable only through their interaction with other systems.” But Heisenberg thought there was a quantum world—just one that operated differently from our own. “The atoms or the elementary particles are not as real [as phenomena in daily life]; they form a world of potentialities or